

Pratik Marudwar

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PROFESSIONAL SUMMARY

Data Scientist with **2+ years** of experience building and deploying **AI/ML solutions** across **healthcare and fintech**. Experienced in **LLM applications** including **AI agents**, **RAG pipelines**, and **Text-to-SQL**, along with **transformer-based NLP** and model fine-tuning using **BERT**, **RoBERTa**, and **LoRA/QLoRA**. Comfortable taking AI solutions from data preparation and development through evaluation, deployment, and monitoring. **M.Tech in Computer Science (Specialization: AI)**, IIT Patna.

SKILLS

Languages & Data: Python, R, SQL, NumPy, Pandas, PySpark, BigQuery, PostgreSQL
LLM & Agentic Frameworks: LangGraph, LangChain, CrewAI, MCP, Anthropic SDK, OpenAI SDK
RAG & Vector Stores: RAG, Text-to-SQL (NL2SQL), ChromaDB, pgvector, Pinecone, FAISS
NLP & Fine-Tuning: PyTorch, Hugging Face Transformers, BERT, RoBERTa, LoRA/QLoRA, DAPT, SFT, Quantization
ML & Statistical Modeling: Linear Regression, Logistic Regression, GLM, Propensity Score Matching, ANCOVA
AI Infrastructure & Serving: vLLM, Ray Serve, FastAPI, Django, Docker, Kubernetes, AWS, GitHub Actions, REST APIs
Evaluation & Observability: RAGAS, LangSmith, LLM-as-a-Judge, human-in-the-loop
Computer Vision: OpenCV, CNNs, Image Classification, Object Detection (YOLO), Vision Transformers (ViT), CLIP

EXPERIENCE

WebMD Associate Data Scientist	Dec 2024 – Present Remote, India
- Built Prescriber Lens, a six-agent LangGraph NL2SQL pipeline that converts pharma sales queries into validated BigQuery SQL, using intent understanding, schema discovery, ChromaDB few-shot retrieval, SQL generation, SQLGlott validation, and human approval; reduced manual query effort by 60% with 85%+ accuracy. - Built an LLM evaluation and observability layer to track query accuracy, SQL validation failures, retry rates, latency, and token/cost usage, enabling systematic evaluation and iterative improvement of the multi-agent pipeline. - Designed campaign ROI measurement pipelines (Propensity Score Matching, ANCOVA, time-aligned pre/post analysis) on large-scale HCP prescription data in BigQuery and R, and automated the measurement pipeline end to end — cutting turnaround by 60% (5 days to 2). - Built an automated geospatial clustering pipeline generating HCP geodomains using hierarchical distance-based clustering, eliminating manual scripting and repeat analyses. - Measured the effectiveness of HCP marketing campaigns by building treatment-control cohorts using Propensity Score Matching, GLM/Logistic Regression, exact matching, and caliper-based matching.	
Rakuten India Trainee Data Scientist	May 2024 – Dec 2024 Bengaluru, India
- Fine-tuned Japanese LLMs for short-text transaction classification and shipped production inference APIs adopted by 5 Rakuten business units. - Performed domain-adaptive continued pretraining (DAPT) of RoBERTa on Rakuten merchant and financial text corpora in PyTorch, then applied supervised fine-tuning for merchant name classification — improving accuracy from 88.05% to 91.35%.	

RESEARCH

Healthcare Complaint Mining and Classification M.Tech Thesis, Indian Institute of Technology, Patna	Jul 2023 – Apr 2024
- Curated a 10,000-record annotated dataset via web scraping and structured human annotation across 9 complaint categories. - Fine-tuned BERT-based transformer models in PyTorch for multi-label complaint classification, achieving 82.95% accuracy on binary complaint detection.	

EDUCATION

Indian Institute of Technology, Patna Master of Technology, Computer Science (Specialization - Artificial Intelligence)	CGPA: 8.02/10 Jul 2022 – May 2024
Maharashtra Institute of Technology, Pune Bachelor of Technology	CGPA: 8.24/10 Aug 2016 – Nov 2020